Task 6 LSTM/RNN Predictive Modelling Report

A recurrent neural network with LSTM gates was implemented to test whether sequence-based modelling could reduce the prediction error for temperature and PM10.

Dataset Summary

- Source file: task6_weather_pm10_hourly_master.csv
- Dataset rows: 24,096
- Dataset range: 2023-03-17 11:00:00 to 2026-05-01 00:00:00
- Testing period: 2026-04-16 00:00:00 to 2026-05-01 00:00:00
- Testing rows: 361

LSTM/RNN Metrics

Model	Target	MAE	MSE	RMSE	R2
LSTM/RNN	temperature	0.498	0.412	0.642	0.980
LSTM/RNN	pm10	1.621	5.322	2.307	0.774

Methodology

- The same merged hourly weather and PM10 source CSV was used.
- The latest 15 days were reserved for testing.
- A 24-hour lookback window was used so each prediction was based on recent sequential patterns.
- Separate LSTM/RNN models were trained for temperature and PM10.
- Each model used all available feature columns except its current target column.
- Feature engineering included cyclical time features, wind-direction sine/cosine values, one-hour lags, 24-hour lags, and rolling averages.

Metric Definitions

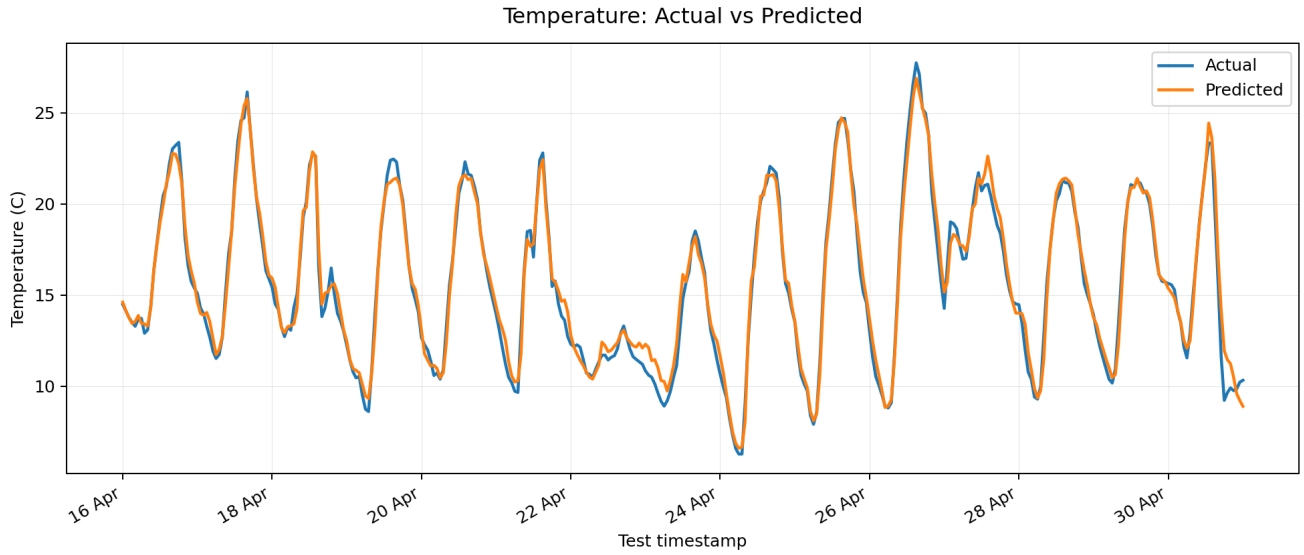
- MSE is the average squared prediction error. Lower values indicate smaller errors, but the unit is squared.
- RMSE is the square root of MSE. It is in the same unit as the target variable.
- R2 measures how much variation in the target is explained by the model. A value closer to 1 indicates a stronger fit.

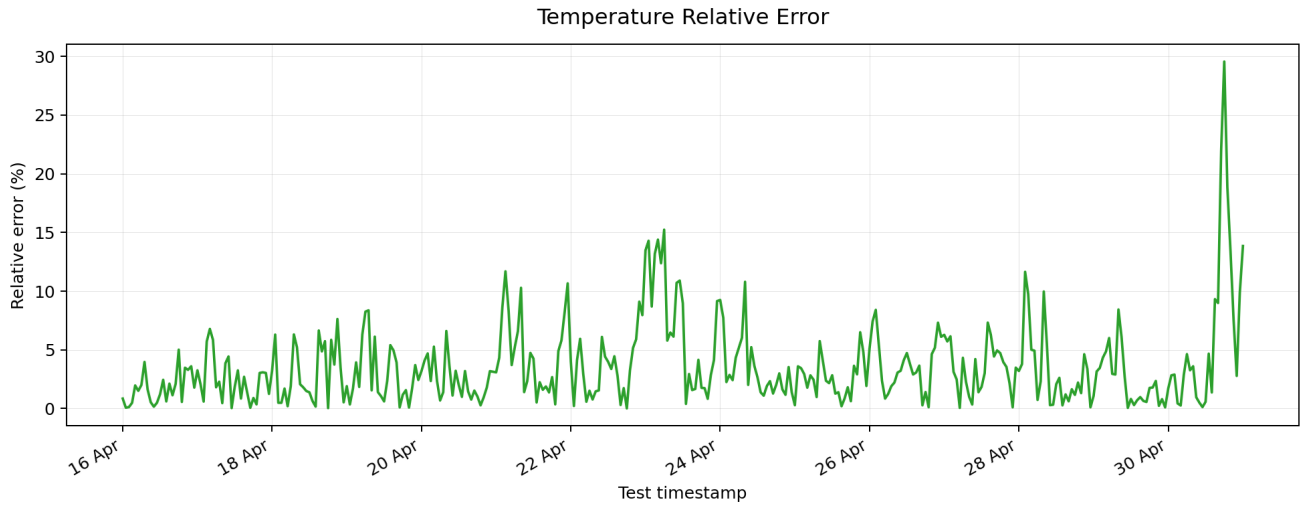
Comparison With Previous Model

Target	Model	MAE	MSE	RMSE	R2
temperature	Gradient Boosting	0.368	0.266	0.515	0.987
temperature	LSTM/RNN	0.498	0.412	0.642	0.980
pm10	Gradient Boosting	1.529	5.286	2.299	0.776
pm10	LSTM/RNN	1.621	5.322	2.307	0.774

The LSTM/RNN model produced a close PM10 result but did not clearly reduce the error below the previous Gradient Boosting model. The result shows that sequence learning was useful to test, but the previous tree-based model remains stronger for this dataset and split.

Temperature Prediction





PM10 Prediction

